# Lab Report: Text classification with neural networks

## 1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. All experiments were run on a single GPU with 24 GB of memory.

## 2 Results

Figure 1 shows the validation accuracy by dataset.

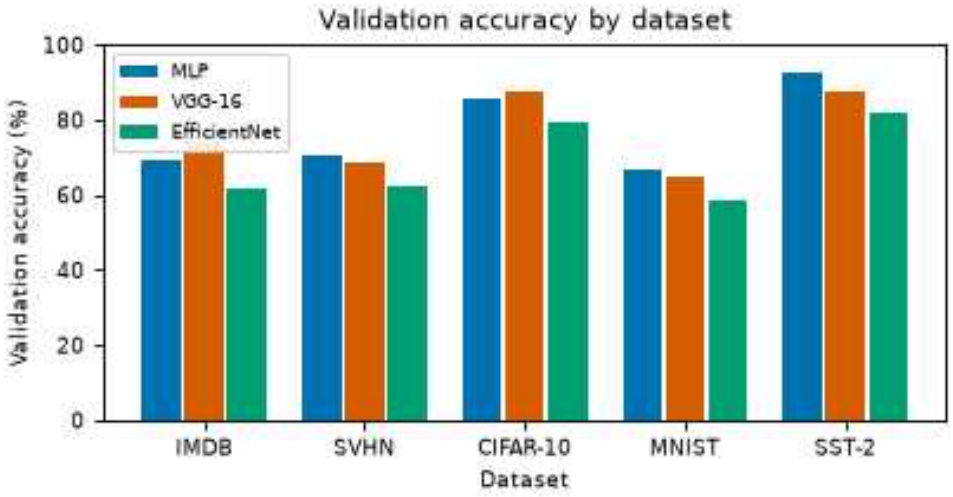



Figure 1: Validation accuracy by dataset for the models in this experiment.

On CIFAR-10, MLP obtains a validation accuracy of 95.9%. On IMDB, VGG-16 obtains a validation accuracy of 74%. MLP outperforms EfficientNet by 7.7 percentage points on SVHN. On IMDB, VGG-16 is 4.6 points better than MLP.

Figure 2 shows the accuracy of ResNet-50 by dataset.

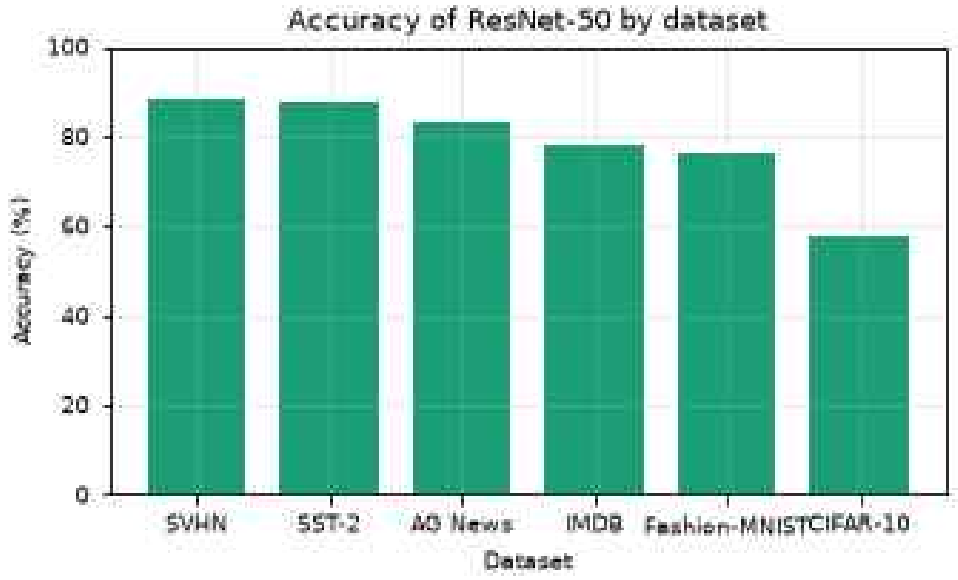



Figure 2: Accuracy of ResNet-50 by dataset for the models in this experiment.

ResNet-50 scores 25.6 points higher on AG News than on IMDB. As shown in Figure 2, on AG News, ResNet-50 obtains a accuracy of over 80%.

## 3 Discussion

On SST-2, XGBoost obtains an accuracy of 90.8%. The differences between the models are discussed in the

next section.